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Paleomagnetism of the Upper Devonian reef complexes, Canning Basin, Western Australia

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Abstract. Paleomagnetism of the Upper Devonian reef complexes in the Canning Basin, Western Australia, was reinvestigated with a primary aim of testing its magnetization age. Samples were collected from back reef and reefal limestones, basinal red beds, syndepositional allochthonous blocks, and sediment infillings in syndepositional cracks and voids. A characteristic remanence, similar to that identified in a previous study, was found in six autochthonous sites in outcrop and four diamond drill cores, with a mean direction of declination $D = 27.5°$, inclination $I = -19.9°$ (half angle of 95% confidence cone $\alpha_{95} = 15.2°$). The primary origin of this remanence was demonstrated by both a syndepositional breccia test and an allochthonous block test. The corresponding Late Devonian pole position (62.0°S, 23.2°E with half angle of 95% confidence cone $A_{95} = 14.6°$) overlaps with the previously reported paleopole from this reef complex and other Late Devonian paleopoles from Australia. The evidence indicates that the spread of late Middle to Early Carboniferous paleopoles from Australia is caused by a phase of rapid apparent polar wander, rather than the tectonic discordance of southeastern Australia in relation to the craton.

Introduction

After the redating of the Msissi Norite in Morocco [*Hailwood*, 1974] as Jurassic [*Salmon et al.*, 1986] and the reinvestigation of the Mereenie Sandstone in central Australia [*Li et al.*, 1991a], which put the original results from that formation [*Embleton*, 1972] into question, the results from the Canning Basin reef complexes [*Hurley and Van der Voo*, 1987] became, for a while, the only Late Devonian pole from cratonic Gondwanaland which put central Africa into a polar position in the Late Devonian. This pole position implied a wide ocean between Gondwanaland and Laurussia at that time [e.g., *Hurley and Van der Voo*, 1987].

Although *Hurley and Van der Voo* [1987, 1990] argued that the remanence from the Canning Basin reef complexes is likely to be of primary origin because of the existence of a layer-parallel polarity pattern at a *Frutexites* locality, a younger age for this pole has been suggested by numerous workers [e.g., *Schmidt et al.*, 1986, 1987, 1990; *Kent and May*, 1987; *Li et al.*, 1988, 1990; *Thrupp et al.*, 1991] because (1) the Canning Basin pole (CB1) is about 40° away from the Givetian–Frasnian CV pole from the Comerong Volcanics (375-370 Ma) in southeastern Australia [*Schmidt et al.*, 1986] but is in good agreement with the HG pole from the uppermost Devonian–Lower Carboniferous Hervey Group in southeastern Australia [*Li et al.*, 1988] and (2) multistage diagenesis in the Canning Basin limestones was documented by *Hurley and Lohmann* [1986].

This study was thus designed to reinvestigate the Canning Basin reef complexes with the major aim being to establish the age of characteristic remanence(s). To achieve this purpose, we designed an allochthonous block test, based on the fold and conglomerate tests [*Graham*, 1949], a syndepositional breccia test, and polarity pattern correlation during field sampling. Our samples came from surface outcrops and diamond drill cores.

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Geological Background and Paleomagnetic Sampling

Underlying an area of 430,000 km^2 onshore in Western Australia, the Canning Basin is a large intracratonic Phanerozoic basin between the Kimberley and Pilbara Precambrian cratons. Geological mapping, drilling, and geophysical data have shown that it contains a maximum aggregate thickness of ~18,000 m of Phanerozoic sedimentary rocks [*Towner and Gibson*, 1983], which overlie unconformably the Precambrian basement. The Paleozoic sedimentary succession commences with Ordovician limestone, shale, and sandstone, which pass upward conformably into Silurian(?) to Middle Devonian evaporites and red beds, Upper Devonian platform carbonates and basinal red beds, Lower Carboniferous shelf carbonates and siliciclastics, Upper Carboniferous to Lower Permian marine clastics interbedded with glacial debris, and Upper Permian marine and fluvial sediments. A gentle deformational event affected the Upper Devonian reef complexes and older sediments in the mid-Carboniferous and is correlated with the Alice Springs Orogeny (ASO) in central Australia [*Forman and Wales*, 1981].

Paleomagnetic samples for this study came from the upper Middle to Upper Devonian reef complexes developed along the ~350-km long Lennard Shelf at the northern margin of the onshore Canning Basin (Figure 1a). The maximum thickness of the reef complexes is ~2000 m [*Playford*, 1984]. Three facies are recognized as follows: platform, marginal slope, and basin. Whereas limestone dominates the first two facies, terrigenous clastics dominate the basin facies (Figures 1b and 2).

A total of 393 surface samples from 36 sites was collected from surface outcrops, 220 from the Pillara Limestone and the Nullara Limestone (platform), the Napier Formation and the Sadler Limestone (marginal slope), and the Virgin Hills Formation (marginal slope and basin) and 173 from the neptu-

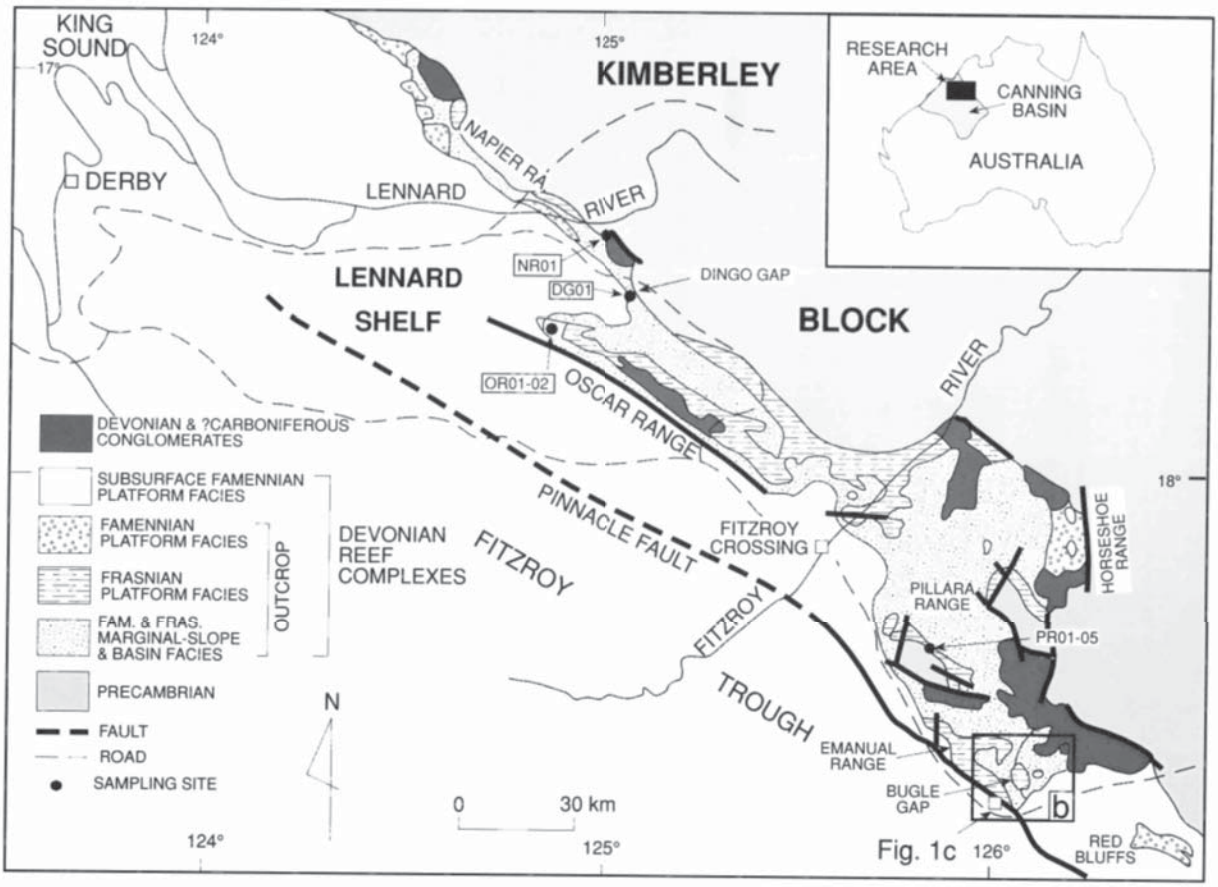


Figure 1a. Geological map of the Upper Devonian reef complexes in the northern Canning Basin and the locations of paleomagnetic sample localities for the whole research area.

nian dykes and sills, geopetal infillings, and allochthonous blocks (Figures 1a, 1b, and 2). The neptunian dykes, sills, and geopetal structures mostly were formed soon after the sediments were deposited in the reef complexes [*Playford*, 1980; 1984] when extensional fractures in the early cemented limestones and other early cavities were filled with sediment filtered down from the surface. These infillings were originally approximately horizontal, so their average attitude can indicate the paleohorizontal at the time of their formation. This feature offers an important paleohorizontal control for paleomagnetic study in these areas. The allochthonous blocks were formed when sections of the platform margin or upper (reefal) marginal slope collapsed along neptunian fractures [*Playford*, 1984] and slid or rolled down the marginal slope. The size of the blocks ranges from a few meters to more than 100 m in long dimension.

Apart from surface samples, 51 samples of reddish siltstone were collected from seven diamond drill cores in the Virgin Hills Formation in the Cadjebut area (Figure 1c). All the diamond drill cores were drilled vertically or semivertically (<4° tilt) by Broken Hill Proprietary Limited (BHP). Drill core locations, orientations, and stratigraphical logging records were supplied by BHP. The dip azimuths of the rock strata cut by cores CD88, CD97, CD98, and CD101 were estimated by correlating the top surface of the Virgin Hills Formation intersected by 12 diamond drill cores in that area. The strike of the nearby regional bedding was used for cores ID535 and ID536. True dips of the strata were calculated from the core orientations, bedding strikes, and the bedding-to-core-axis angles. Samples from these six cores were orientated by first restoring the cores to their original attitude and then by rotating the samples around the core axis, so as to align the dip azimuth of the bedding plane identified on each sample with that of the estimated regional bedding plane. The orientation of samples from cores CD96 and CD99 was achieved by rotating the samples around the core axis until the Holocene overprint component identified from the cores pointed to the north.

Estimate of Tilt Angles

Although significant primary dip exists in the reef complex (particularly in the fore reef facies), tectonic tilting is negligible for most of the localities. Syndepositional geopetal fabrics and in situ organic growth forms have been used as paleohorizontal indicators [*Playford*, 1980], which show that postdepositional tilt (mostly due to differential compaction) for the autochthonous localities is minor (0-15°). At three sites (Bugle Gap localities BG04-06) from western Ngumban Cliff (Figure 1b) and the seven drill cores from near Cadjebut (Figure 1c) there are no geopetal measurements available. A 6° tilt correction was applied to samples from the western Ngumban Cliff for two reasons. The first is that postdeposi-

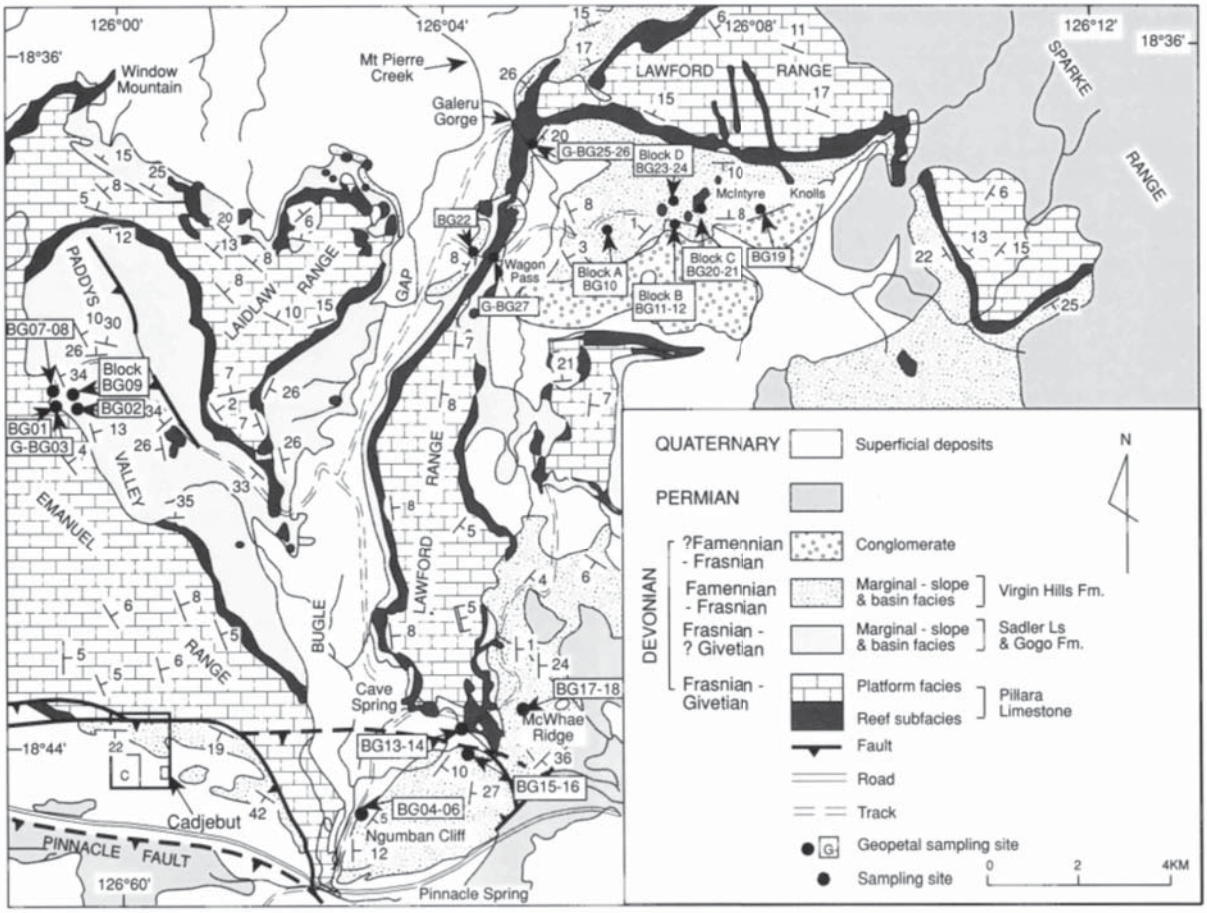


Figure 1b. Geological map of the Bugle Gap region [after *Playford*, 1981, 1984].

tional steepening of dip ranges 0° to 10° in the Bugle Gap area, as indicated by geopetal measurements at several localities, such as McWhae Ridge (7°), western Paddy Valley (0°), Wagon Pass (10°), Galeru Gorge (4°), and McIntyre Knolls (3°–7°). The second is that the dip of the regional bedding for BG04–6 is only 6°. A tilt correction to account for a regional tilt of the paleohorizontal by 15° toward 022° has been applied to sites PR01–03 from the platform facies in the Pillara Range (Figure 1a) because the bedding there dips about 20° toward the north-northeast and geopetal structures in the adjacent fore reef deposits tilt at 15° towards 022°. For the drill cores (Figure 1c), where almost all the cores have apparent dips around 10° (with core CD96 being the only exception with an apparent dip of 24°), an arbitrary 7° tilt correction to shallow the dip has been applied, as these cores were drilled in the basinal facies which were originally fairly flat lying and less affected by the differential compaction. For the allochthonous blocks the samples can only be restored to their paleohorizontal using geopetal orientations but without strike (declination) control, owing to their unknown rotation and rolling histories.

Age Constraint

The whole complex ranges from late Givetian to late Famennian in age. Detailed paleontologic and biostratigraphic study on the Devonian of the Canning Basin is based mainly on ammonoids [*Glenister*, 1958; *Petersen*, 1975], conodonts [*Glenister and Klapper*, 1966; *Seddon*, 1970; *Druce*, 1976; *Nicoll and Druce*, 1979], corals [*Hill and Jell*, 1970], bryozoans [*Ross*, 1961], brachiopods [*Coleman*, 1951; *Veevers*, 1959; *Grey*, 1978], algae [*Wray*, 1967], stromatolites [*Playford and Cockbain*, 1969; *Playford et al.*, 1976], plant microfossils [*Balme and Hassell*, 1962; *Grey*, 1974; *Playford*, 1976], and fish [*Long*, 1988]. The faunal distribution within the complexes has been outlined by *Playford* [1980, 1984]. The stratigraphic location and facies of each paleomagnetic sampling site are shown in Figure 2. All these sites are within an age range of late Frasnian–early Famennian (~365 Ma).

Paleomagnetic Results and Interpretation

The intensities of natural remanent magnetization (NRM) of reddish limestones and siltstones are mostly within a range of 0.5 to 5 mA m^{-1} (x 10^{-6} emu/cm^3), that of gray limestones between 0.4 to 1 mA m^{-1}, and that of geopetal infillings from 0.2 to 8 mA m^{-1}. NRM intensities of about half of the surface samples are around 1 mA m^{-1} or lower. Remanence of these samples was thus measured using a three-axis CTF cryogenic magnetometer at the Commonwealth Scientific and Industrial

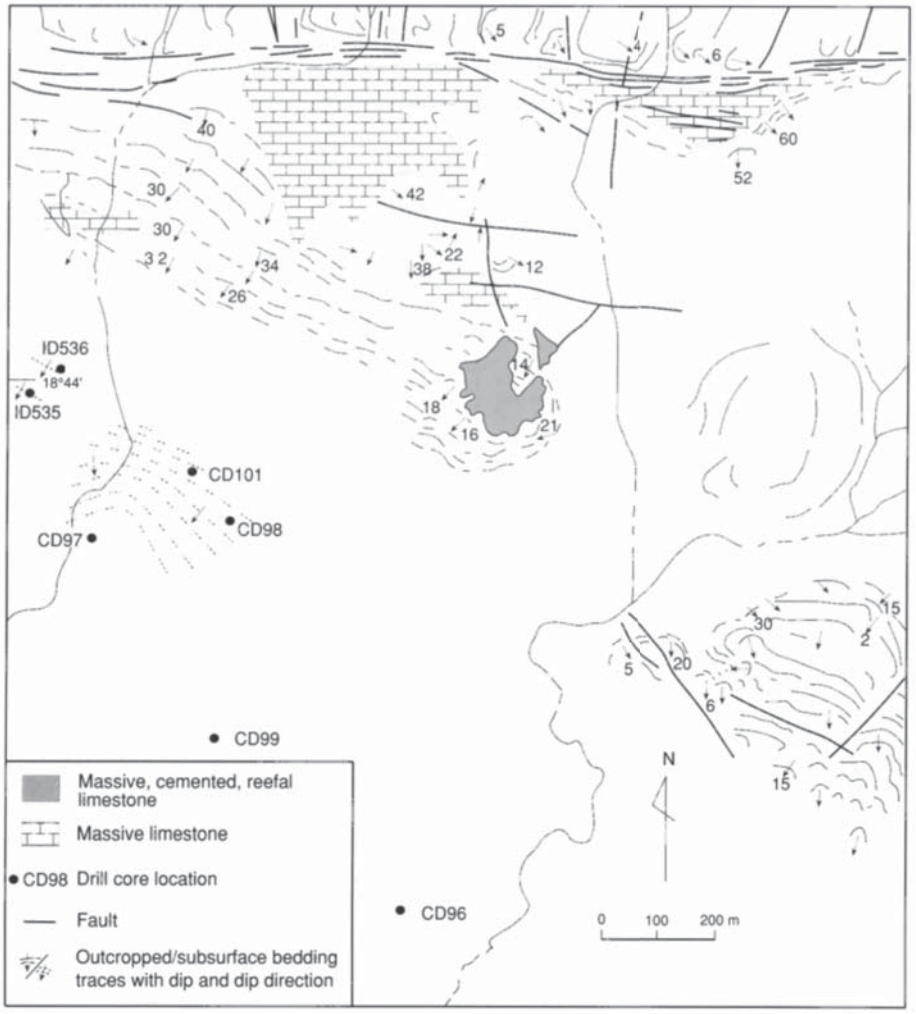


Figure 1c. Regional geology and locations of the studied drill holes at Cadjebut (see inset in SW corner of Figure 1b) [after Broken Hill Proprietary Limited, unpublished map].

Research Organization (CSIRO) Rock Magnetism Laboratory in Sydney. Remanence of the rest of the samples was measured using an upgraded Digico spinner magnetometer at the Paleomagnetism Laboratory of the University of Western Australia (UWA).

All samples were demagnetized stepwise. Samples measured at CSIRO were demagnetized using a three-stage autocontrolled carousel furnace installed within a zero-field space (≤5 nT), and those processed in the UWA laboratory were demagnetized using both a modified Schonstedt TSD furnace and a Molspin two-axis tumbler AF demagnetizer. The principal component analysis technique [*Kirschvink*, 1980] and orthogonal plots [*Zijderveld*, 1967] were employed interactively using the CSIRO ROCMAG package for data analysis. Linearity, which is defined as the ratio of the distance between the bounding data points to the sum of the distance between the intervening points [*Schmidt*, 1982], was used for automatic searching of linear segments in individual specimens. Most of the selected linear segments have a linearity between 0.9 and 1. For final analysis, sites (or cores) with less than three samples carrying the same remanent component or having the half angle of its 95% confidence cone (α_{95}) ≥30° were excluded from the mean direction calculations.

Two characteristic remanent components have been identified, a Holocene geomagnetic field overprint (C1), and a primary remanence (C2) (Table 1). A sporadic remanent component (C3), for which no geological significance can be determined, was also found in three sites (Table 2).

Recent Geomagnetic Field Overprint

This remanence (C1, Figure 3) is well developed in both surface and diamond drill core samples. Its characteristic direction is inclined moderately up to the north (Figure 4) with a mean direction of declination D = 0.9°, inclination I = -49.4° (number of sites N = 26, α_{95} = 2.9°, precision parameter k = 110.8). It is resistant to AF cleaning (Figure 3a) but is easy to remove by low-temperature thermal demagnetization (Figures 3a, 3b, and 3f). The unblocking temperature of C1 is generally lower than 350°C but is over 575°C in some reddish limestone and siltstone samples.

The in situ site mean directions of C1 group well and are close to the present geomagnetic field direction (Figure 4) but

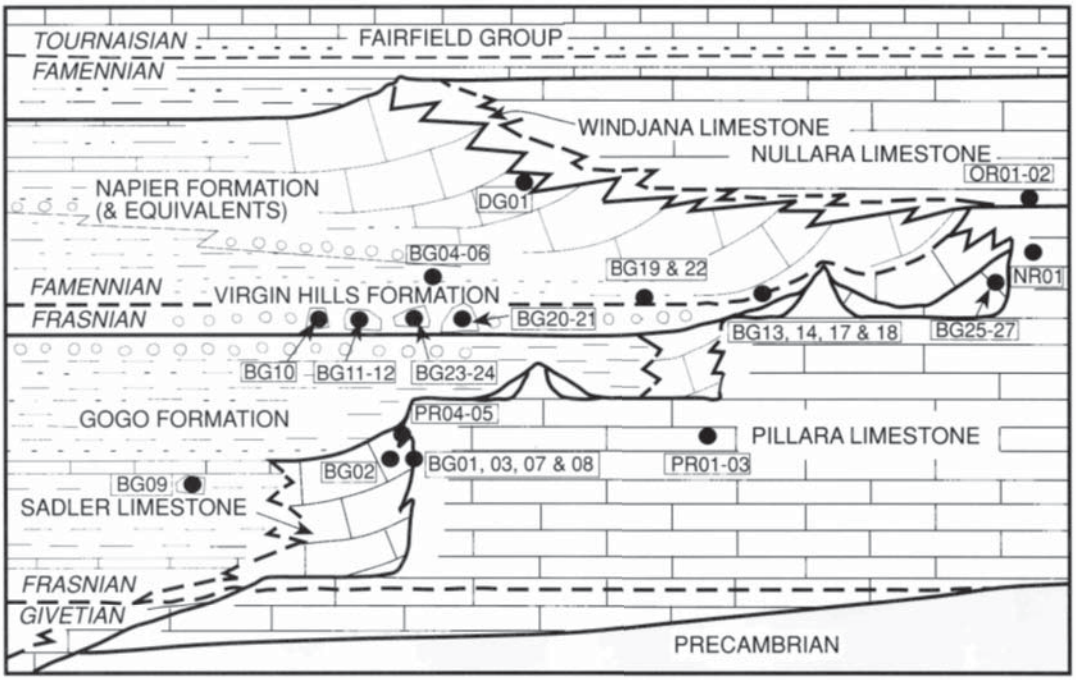


Figure 2. Schematic stratigraphic section showing age and facies of palaeomagnetic sample localities [after *Playford*, 1984].

become significantly dispersed after tilt correction (parameter k changes from 110.8 to 11.1). It therefore most likely represents a Holocene geomagnetic field overprint.

Late Frasnian–Early Famennian Primary Remanence

A remanence characterized by moderate to high unblocking temperature and shallow inclination, directed either upward toward north to northeast (C2 in Figures 3c–3g) or downward to southwest (C2, Figure 3d), was isolated from 33 samples in six sites (BG05, BG14, BG17; and Oscar Range localities OR01, OR02, and PR01) and 18 samples from four drill cores (CD96, CD98, CD99, and CD101). Sites OR01, OR02, and Pillara Range locality PR01 are from the platform limestone facies; sites BG14 and BG17, from the marginal slope facies; and site BG05 and the four drill cores, from basin facies (Figures 1 and 2). Vectors of this remanence decay approximately toward the origin (Figure 3). Although AF demagnetization has been partially effective in removing this remanence (Figure 3e), thermal demagnetization is demonstrated to be more effective. The maximum unblocking temperature for samples from sites BG14 (reddish siltstone) and PR01 (gray limestone) is in the range of 530°C to 560°C, and for samples from other sites and diamond drill cores (reddish siltstone and limestone) it is up to 690°C. Therefore both magnetite and hematite are inferred to be the contributors to this remanence.

The site mean directions are given in Table 1 and Figure 5. Neither the directions nor their groupings change much by the gentle tilt correction. In order to examine the age of this remanence, three other field tests have been carried out.

Syndepositional breccia test. The syndepositional breccia test used follows the principle of the conglomerate test. Ten samples from site BG05 were collected from a 2-m thick succession with syndepositional breccia (Figure 6a) [*Kerans and Playford*, 1984; P. E. Playford, personal communication, 1992] near western Ngumban Cliff. The succession can be divided into three units, a relatively undisturbed lowermost bedding unit (unit 1), a syndepositional breccia (unit 2) and an uppermost white limestone (unit 3) (Figure 6a). Stylolites are common in the syndepositional breccia. Seven samples, A to G, are reddish siltstone from units 1 and 2, and the other three are white limestone from unit 3.

Only one stable, high unblocking temperature (650°C to 690°C) component (Figures 6b and 6c) was found in each sample. Hematite appears to be the major magnetic carrier. The magnetic directions of all samples are plotted in Figure 5a. Directions of samples from unit 1 (A to D) group fairly well, and the inclination angles of these directions are shallower than the Holocene field overprint. However, two of the three samples from unit 2 (E-G) have their remanent directions ~90° away from the group of directions in unit 1. Magnetic directions in the white limestone (unit 3), which could be a younger neptunian sill, are also divergent from the group of directions in unit 1. This phenomenon of remanent directions being discordant between disturbed and nondisturbed units suggests that no remagnetization event has occurred here since the formation of the syndepositional breccia. Therefore the shallow upward-to-north direction in samples from the least disturbed unit (unit 1) (Figures 5a and 6b) should be regarded as the Late Devonian characteristic direction of the Canning Basin.

Allochthonous block test. The allochthonous block test basically follows the principle of a conglomerate test, with the additional constraint that because there is paleohorizontal information for each allochthonous block, there is also

Table 1. Primary Remanence (C2) Directions

Site/ Core	Location °N	Location °E	Tilt Corrected	n/n_0 (N)	In Situ D, deg	In Situ I, deg	100% Untilted D, deg	100% Untilted I, deg	α_{95}, deg	k	Pole Position Plat, deg	Pole Position Plong, deg
Surface Samples												
BG05	-18.75	126.05	6/232[a]	5/10	349.9	-28.9	352.6	-26.0	22.7	12.3	-81.7	250.7
BG14	-18.73	126.07	7/190[b]	6/11	012.7	-20.5	012.6	-13.5	20.4	11.7	-72.9	353.7
BG17	-18.72	126.09	7/113[c]	6/10	-47.5	-26.7	044.1	-29.4	11.0	38.2	-47.9	039.1
OR01	-17.65	124.93	7/220[a]	6/16	223.6	23.9	223.4	16.9	27.9	06.7	-46.9	028.6
OR02	-17.65	124.93	10/284[a]	5/10	200.6	40.4	208.7	38.5	23.8	11.3	-62.7	048.1
PR01	-18.40	125.83	15/022[b]	5/8	049.3	-14.8	052.1	-28.0	12.8	36.6	-40.1	040.0
Mean of 6 sites				(6)	028.1	-27.5	029.0	-26.9			-61.7	032.1
α_{95}						19.6		19.0				
A_{95}												19.1
k						12.7		13.4				
K												13.2
Normal				22			025.2	-29.0	10.8	9.2		
Reverse				11			217.9	20.7	18.9	6.8		
Diamond Drill Core Samples												
CD96	-18.75	126.00	7/174[c]	5/8	240.8	16.5	239.1	13.6	30.0	7.5	-31.4	032.6
CD98	-18.73	126.00	7/218[c]	5/8	001.2	-11.8	001.9	-6.2	22.8	12.2	-74.3	313.0
CD99	-18.73	126.00	7/200[c]	5/8	004.7	-11.5	005.0	-4.8	28.9	07.9	-73.0	323.3
CD101	-18.73	126.00	7/201[c]	4/8	037.8	-15.8	037.4	-9.1	22.2	18.1	-50.9	19.6
Mean of 4 cores				(4)	025.6	-15.2	025.4	-9.2			-61.3	009.8
α_{95}						32.8		32.4				
A_{95}												32.5
k						8.8		9.0				
K												9.0
Normal				12			019.6	-10.6	13.9	10.8		
Reverse				7			215.0	8.6	39.6	3.3		
Overall Mean of This Study												
				(10)	027.1	-22.7	027.5	-19.9			-62.0[d]	023.2[d]
α_{95}						14.9		15.2				
A_{95}												14.6
k						11.4		11.0				
K												11.9
Normal				34			023.1	-22.5	8.7	8.9		
Reverse				18			216.8	16.4	17.2	5.0		
Combined Results of Hurley and van der Voo [1987] and This Study												
				(17)			033.7	-24.0	9.6	14.9	56.7	030.9
A_{95}												9.0
K												16.6

Abbreviations are n/n_0, number of samples for mean calculation/total sample number; (N), number of sites (or cores); D, declination; I, inclination; α_{95}/A_{95}, half angle of the 95% confidence cone for mean direction/paleopole position; k/K, precision parameter of *Fisher* [1953] distribution for mean directions/pole positions; Plat, pole latitude; Plong, pole longitude.

[a] Value indicates bedding measurement.

[b] Value indicates geopetal dip and bedding strike.

[c] These values are estimates.

[d] Values used as south paleopole (CB2) in Figure 9.

an element of a fold test [*Graham*, 1949]. Each allochthonous block is treated as a large clast within the host marginal slope deposits. Geopetal infillings develop before (early) and after (late) the allochthonous blocks are dislodged from the platform or upper marginal slope. Paleomagnetic samples were taken from both the early and late geopetal infillings, and the paleomagnetic remanence of the early infillings are used in this test. Four separate allochthonous blocks at the McIntyre Knolls and one in the western Paddys Valley (Figure 1b) were sampled.

Ninety-four geopetal samples were collected from the four blocks in the McIntyre Knolls (Figures 7a–7d), 56 from the

Table 2. Remanence Directions (C3) With Undetermined Origin

Site/ Core	Location °N	Location °E	n/n_0	In Situ D, deg	In Situ I, deg	100% Untilted D, deg	100% Untilted I, deg	α_{95}, deg	k
BG17	-18.72	126.09	4/10	310.9	-34.2	309.7	-27.5	09.1	102.7
PR03	-18.40	125.83	7/8	320.3	15.2	323.0	7.7	20.3	9.8
PR05[a]	-18.37	125.82	6/11	313.4	23.7	319.8	22.4	25.8	7.7

Abbreviations are same as in Table 1.
[a] Site from geopetal infillings.

early geopetal infillings in four blocks and 38 from the late geopetal infillings in three blocks. The early geopetal infillings are commonly brightly colored with red and yellow alternating bands, in contrast with the late geopetal infillings which are a duller salmon pink. Early geopetal orientations were measured in all four blocks and late geopetal structures in three of the four blocks (blocks B–D). In addition, bedding defined by limestone layers was measured in three of the blocks and was found to be statistically indistinguishable from the paleohorizontal given by the early geopetal structures (Figures 7b–7d).

The mean orientation of the early geopetal structures was used to define the original paleohorizontal in the allochthonous blocks. The considerable scatter in the early geopetal orientations within each block (Figures 7a–7d) is caused by two factors (1) primary dip of geopetal sediment layers due to the confined and irregular shape of many of the cavities and (2) partial fragmentation and rotation of the blocks during emplacement. Nonetheless, the parallelism of mean orientation of early geopetal structures with the bedding supports the interpretation that the mean geopetal attitude is a good estimate of the paleohorizontal and also indicates that the blocks have been derived mainly from a part of the reef complex where the bedding is subhorizontal.

Apart from the Holocene field overprint, characteristic remanence was revealed from the early geopetal infillings in three blocks (Figures 7e, 7g, and 7h) and from the late geopetal infillings in one block (Figure 7f). The characteristic remanence from blocks A, C, and D shows considerable scatter both within and between each block (solid and open circles in Figures 7i, 7k, and 7l). However, when the remanent directions are restored to paleohorizontal by rotating the mean geopetal orientation in each of the three blocks to horizontal, they have a shallow inclination (triangles in Figures 7i, 7k, and 7l). This suggests that the characteristic remanence in both the early geopetal infillings and the autochthonous strata was acquired before the blocks were incorporated in the marginal slope. This test gives an extremely tight time constraint for the characteristic magnetization, as the blocks were emplaced during sedimentation.

The scatter of remanence directions within each block can be accounted for partly by normal fluctuation about the mean and partly by fragmentation within the blocks during emplacement. The difference in remanence orientation between blocks is to be expected because these blocks could have rotated by unknown amounts in the horizontal plane since dislodgment from their original depositional sites. There is thus no control on the declination of the primary remanence within the blocks. One interesting point is that in the four large blocks measured at McIntyre Knolls the early geopetal structures are inclined at ~60° to the enclosing marginal slope bedding, suggesting that all four blocks have basal sliding surfaces on reef fractures originally inclined at ~60° to the paleohorizontal. The blocks have also rotated, so that the stratigraphic tops face downslope, as might be expected if the blocks were wedge shaped with the originally 60° failures the largest surfaces.

The remanence from the late geopetal infillings in block B has steep inclination but is quite scattered (Figure 7j). These directions are different from those in the early geopetal infillings and closer to the mid- to Late Carboniferous overprint reported in other parts of Australia [e.g., *Li et al.*, 1989; *Idnurm and Heinrich*, 1993; *Klootwijk et al.*, 1993], suggesting that infilling of the geopetal cavities could have occurred at that time.

Thirteen geopetal samples were collected from an allochthonous block (site BG09) in the late Givetian to early Frasnian Sadler Limestone in the western Paddys Valley (Figure 1b). The block contains bedding at a variety of attitudes, though it is not clear whether this variation was acquired during block emplacement or later. Both AF and thermal demagnetization were sufficient to clean the NRM in all the samples. After restoration of the bedding attitude to horizontal almost all the inclinations fall in the range of 1° to ±27°, again suggesting that the remanence in the source rocks, which also has a shallow inclination, should be primary.

Magnetostratigraphy test. Correlatable magnetic polarity patterns were the prime evidence for a primary origin of the characteristic remanence in the study by *Hurley and Van der Voo* [1987]. In this study, dual polarities are present in all but one site and three out of the four drill cores which contain C2 (Figure 5). Samples from two sections in the western Oscar Range, ~200 m apart, which can be correlated by walking out the outcrop, have been used for a polarity test. The two sections, each about 10 m thick, comprise the uppermost Frasnian–Famennian Nullara Limestone disconformably overlying the Pillara Limestone (Figure 8). In this locality the Pillara is a gray massive limestone and the Nullara is gray to salmon pink bioclastic limestone with interbedded fissile limestone.

Twenty-six samples were collected from the two sections (Figure 8). Apart from sample A of site OR01 all the samples were from rock units just above the Pillara/Nullara disconformity. There are 11 samples (six from OR01 and five from

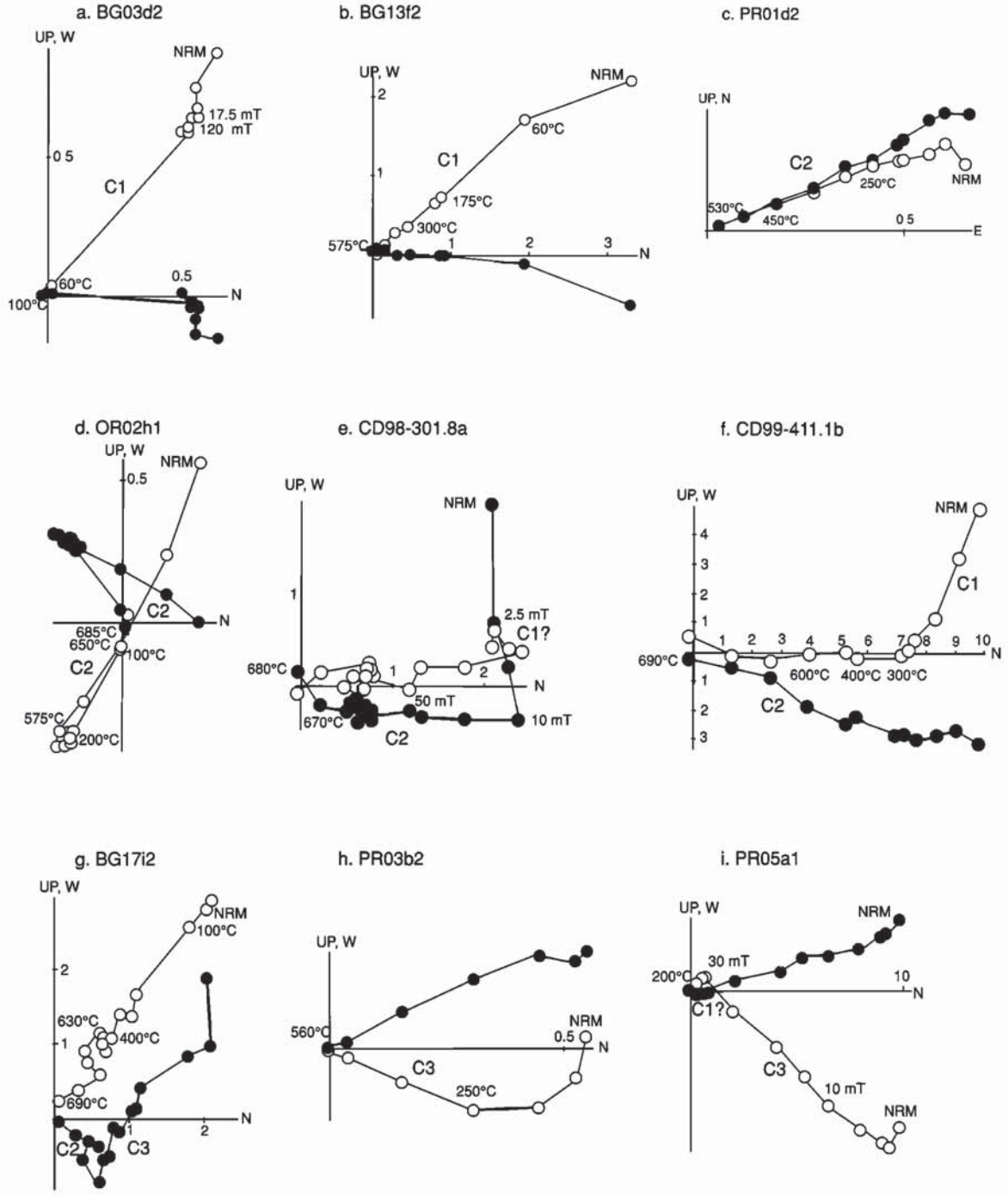


Figure 3. Representative orthogonal plots of the in situ magnetization vectors revealed from samples (a) BG03d2, (b) BG13f2, (c) PR01d2, (d) OR02h1, (e) CD98-301.8a, (f) CD99-411.1b, (g) BG17i2, (h) PR03b2, and (i) PR05a1 from the Devonian reef complexes (excluding geopetal samples). Solid (open) circles are projections on horizontal (vertical) planes. Magnetic components are as follows: C1, Holocene field overprint; C2, primary remanence; and C3, sporadic remanence of unknown origin, found in sites BG17, PR03, and PR05.

OR02) in the Nullara Limestone containing the characteristic remanence. Samples from the bioclastic layers contain only normal polarity remanence, and samples from the gray to salmon pink limestone contain both normal and reversed polarities. Although the limited number of usable samples made the two polarity columns incomplete, the pattern of reversals is consistent with the geological correlation. As with *Hurley and Van der Voo's* [1987] results, this suggests (although not

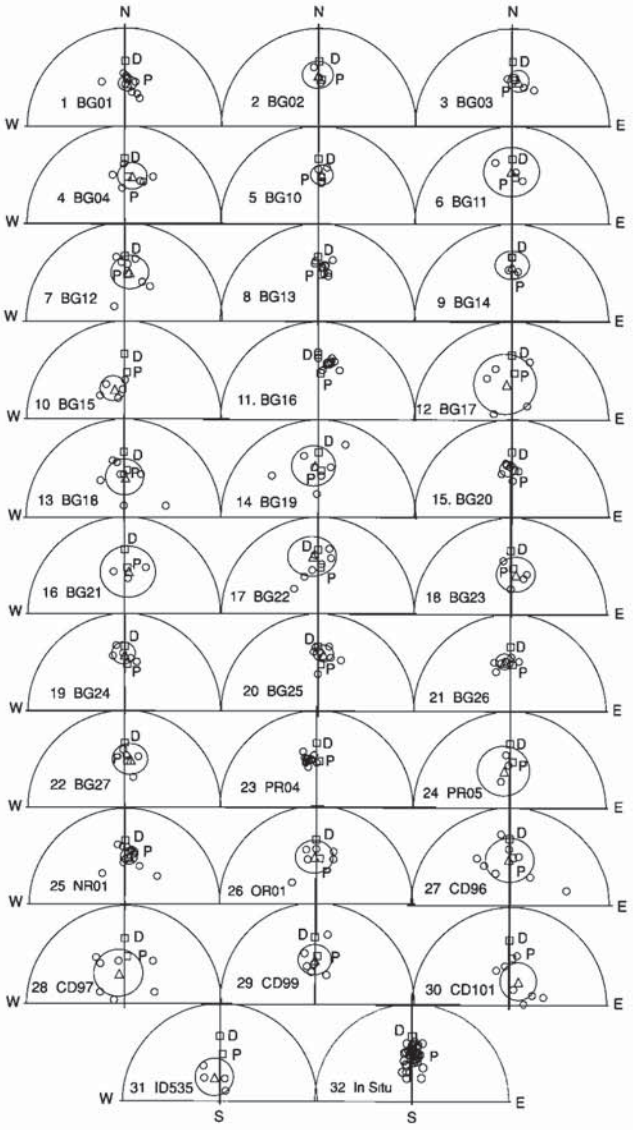


Figure 4. Upper hemisphere equal-area stereographic projections of C1 directions. Numbers 1-31 are in situ (geographic) sample mean directions; 32, core mean directions in in situ position. The means (triangles) and their 95% confidence circles (small circles) are also shown for each site. P is present geomagnetic field direction, and D is present dipole field direction.

very strongly) that the magnetic remanence is likely to be a latest Frasnian-earliest Famennian primary remanence, rather than an overprint.

Summary. All the field tests indicate that no regional remagnetization event has occurred in the studied field area since the Upper Devonian reef complexes were formed. The characteristic remanence, C2, discussed above thus has to be of primary origin, i.e., has an age of late Frasnian to early Famennian (~365 Ma).

The mean direction of the primary remanence from drill cores (inclination -9.2°/declination 025.4° with α_{95} = 32.4° and k = 9.0) agrees with that of the surface samples (-26.9°/029.0° with α_{95} = 19.0° and k = 13.4) (Table 1). The overall mean direction of -19.9°/027.5° with α_{95} = 15.2° and k = 11.0 gives a paleopole (CB2) at (62.0°S, 023.2°E) with A_{95} = 14.6° (Table 1). The total of 52 samples (34 normal and 18 reversed) passed the reversal test of *McFadden and McElhinny* [1990] with a "C" classification.

Remanence With Undetermined Origin

A third possible remanent component was isolated from 21 samples in the three following sites: BG17 from the Virgin Hills Formation at the McWhae Ridge, PR03 from the Pillara Limestone, and PR05 from geopetal infillings in Sadler Limestone, Pillara Range (Figures 1a and 1b and Table 2). Site BG17 is in reddish siltstone, PR03 in gray limestone, and PR05 in brown to reddish cement geopetal structures. The predominant direction of the remanence is either shallow downward or upward to northwest or upward to southeast (Figures 3g-3i). The remanence can be eliminated by either AF demagnetization with a peak field between 30 and 50 mT (300–500

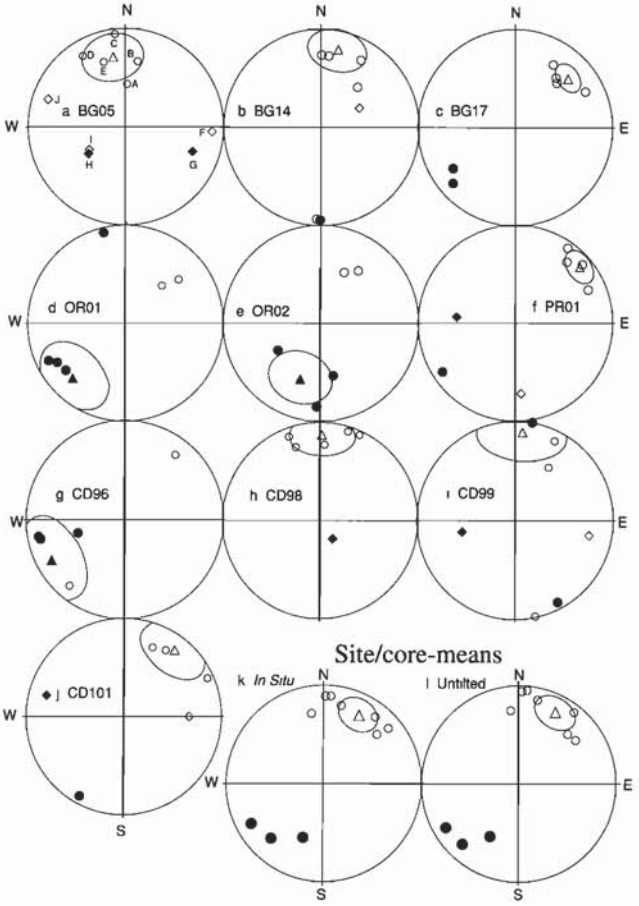


Figure 5. Equal-area stereographic projections of C2 directions. (a)-(j) In situ (geographic) sample mean directions. (k) Core mean directions in in situ position. (l) Core mean directions in paleohorizontal position. Solid (open) symbols are projections on lower (upper) hemisphere. Diamonds are the sample directions which were excluded from core mean direction calculations.

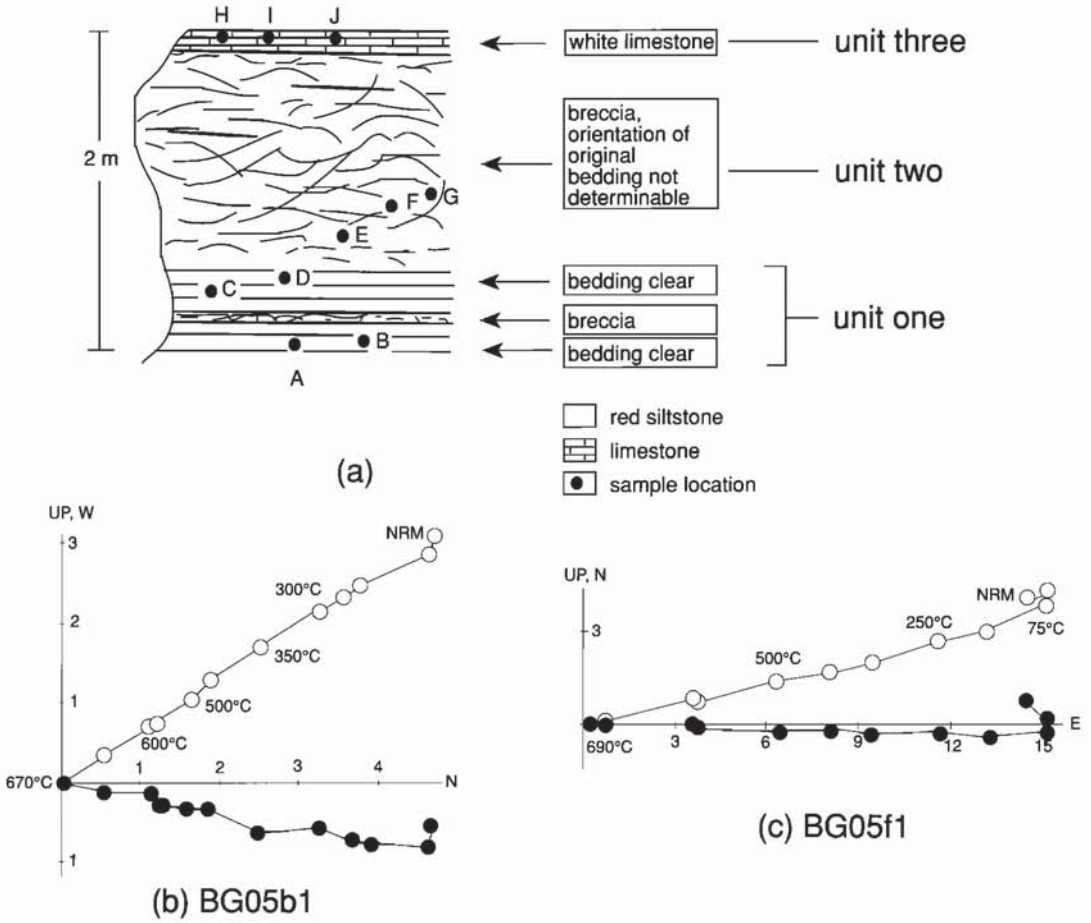


Figure 6. Syndepositional breccia test showing the remanence is of prebrecciation origin. (a) Sampled section at site BG05 in a breccia layer near western Ngumban Cliff, Bugle Gap. (b) Orthogonal plot for BG05b1. (c) Orthogonal plot for BG05f1. Stereographic plot of sample mean directions showing the scattering of directions in the brecciated units is given in Figure 5a.

Oe) (Figure 3i) or moderate- to high-temperature thermal demagnetization (Figures 3g and 3h), with its direction sometimes offset from the origin on orthogonal plots (Figures 3g and 3i).

The origin of this possible remanence is unclear. One possibility is that it represents an overprint event, as suggested by its coexistence with the primary remanence at site BG17 but with a lower unblocking temperature (Figure 3g). However, as discussed earlier, no such overprint has been found in the majority of the sites, including sites where the breccia, block, and polarity tests were conducted. Other possibilities include lightning strike or unsuccessful vector separation.

Discussion

The paleopole corresponding to the primary remanence (CB2) is plotted in Figure 9, together with existing Devonian to Early Carboniferous paleopoles from Australia. All these poles are based on results from at least 12 samples with $\alpha_{95} \leq 30°$ and have good paleohorizontal control. Their quality classifications are C or above, according to the modified *Briden and Duff* [1981] classification scheme [*Li and Powell*, 1993], and $Q \geq 3$, according to the classification scheme of *Van der Voo* [1990].

The main reason which led previous workers to suspect the tectonic/structural coherency of the CV pole from the Comerong Volcanics obtained by *Schmidt et al.* [1986] from southeastern Australia or the primary origin of the CB1 pole previously obtained from the Canning Basin reef complexes by *Hurley and Van der Voo* [1987] is the angular distance of >40° between the two paleopoles. However, as demonstrated by the field tests carried out in this study, the Canning Basin reef complexes have not been pervasively remagnetized since their deposition. Thus the suggestion that the CB1 pole might represent an overprint after the deposition of the reef complexes can no longer be sustained. Furthermore, the 95% confidence oval of the new paleopole obtained in this study (CB2), as well as that of the pole based on the combined results of this study and the *Hurley and Van der Voo* [1987] study, intersect those of Late Devonian poles WP from the Worange Point Formation [*Thrupp*, 1991], HS from the Hermannsburg Sandstone [*Li et al.*, 1991a], HG from the Famennian succession of the Hervey Group [*Li et al.*, 1988], and the newly determined BC pole from the latest Famennian succession of the Brewer Conglomerate [*Chen et al.*, 1993].

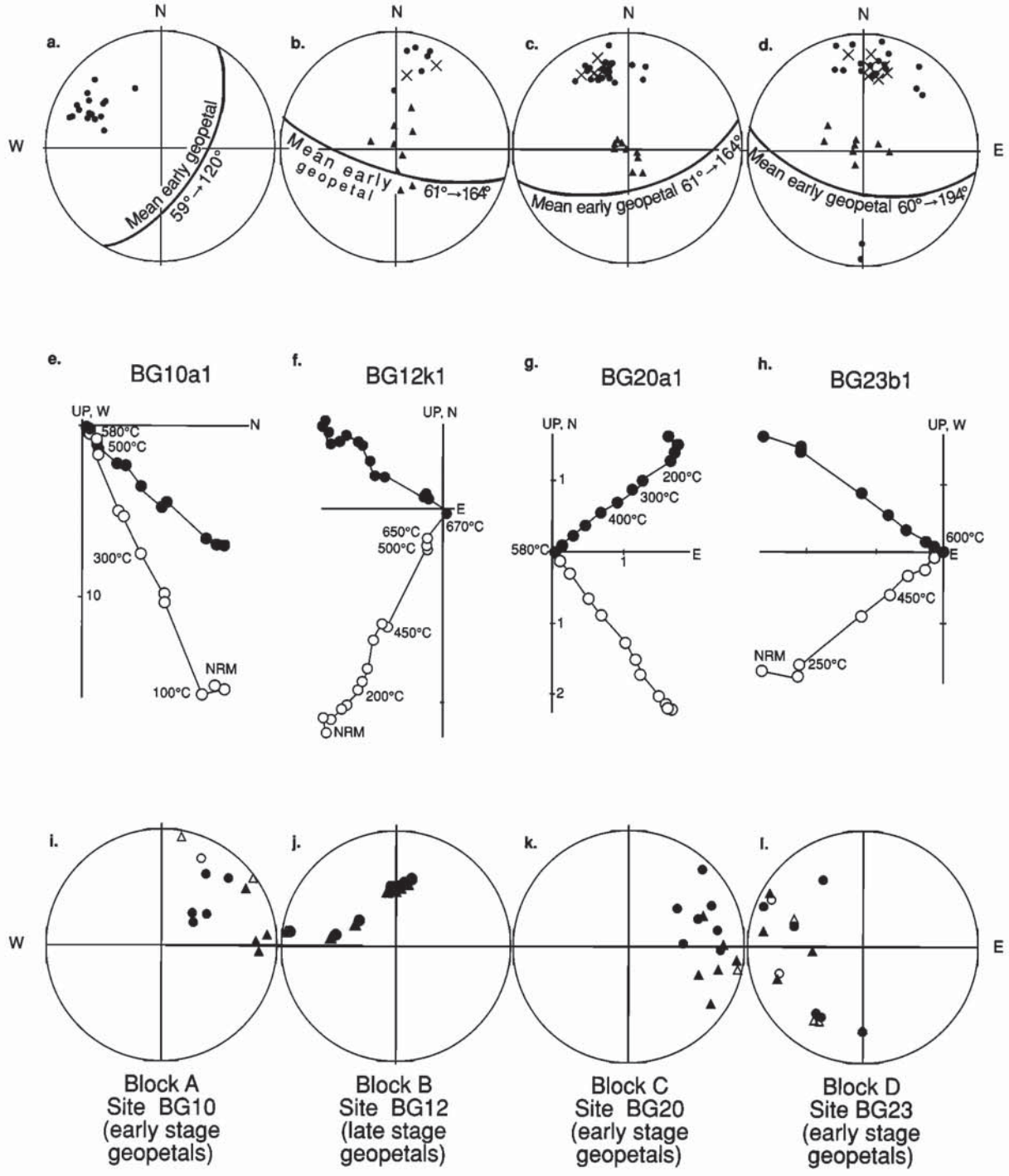


Figure 7. Allochthonous block test from McIntyre Knolls showing the C2 remanence is of primary origin. (a)-(d) Equal-area stereographic projections for blocks A, B, C, and D, respectively, showing orientation of early geopetal infillings (circles), late geopetal infillings (triangles), and original bedding (crosses). (e)-(h) Representative orthogonal plots of the in situ magnetization vectors revealed from geopetal samples, following the same convention as in Figure 3; scales are in 1 mA m^{-1}. (i)-(l) Equal-area stereographic projections of characteristic remanence in blocks A-D, respectively. Circles (triangles) are sample mean directions before (after) tilt correction about the mean early geopetal infilling in Figures 7i, 7k, and 7l and the late geopetal infilling in Figure 7j. Solid (open) circles/triangles are projections on lower (upper) hemisphere.

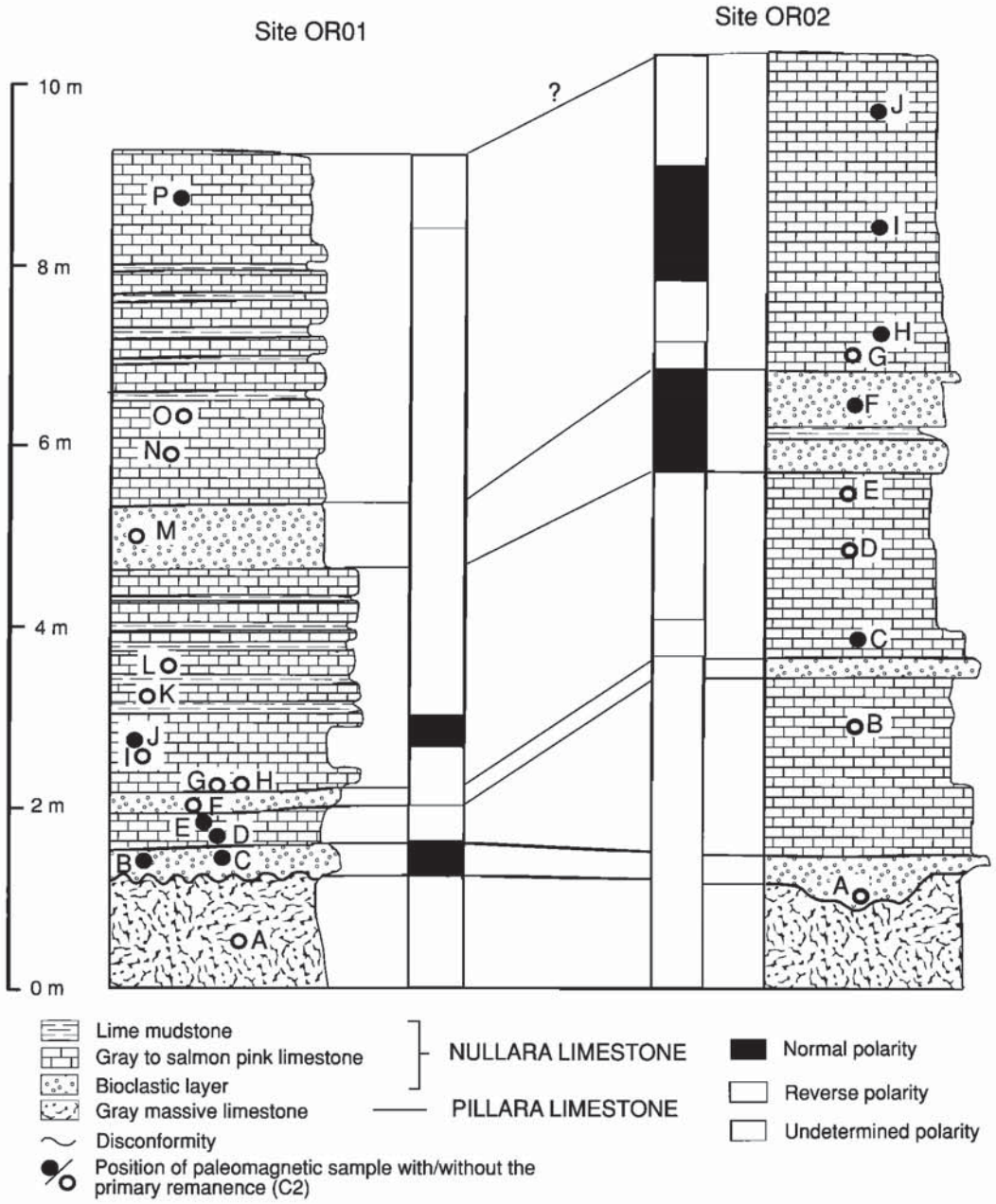


Figure 8. A sketch of sampled sections at sites OR01 and OR02 at the western end of the Oscar Range and their magnetic polarity columns.

Instead of invoking the structural/tectonic discordance of the Comerong Volcanics to account for the difference between pole CV and the Late Devonian poles mentioned above, the spread of Middle/Late Devonian (~375 Ma) to Early Carboniferous (~340 Ma) paleopoles from Australia can be interpreted as caused by a phase of rapid apparent polar wander (APW) [*Chen et al.*, 1993; *Li et al.*, 1993a; *Meert et al.*, 1993]. According to this interpretation, the Gondwanan APW path started from southern South America in the late Middle to early Late Devonian (375–370 Ma, pole CV), "migrated" across eastern South America to the African Bight by the end of the Late Devonian (360–355 Ma, poles WP, HS, CB2, HG, CB1, and BC), and finally reached central Africa by the Early Carboniferous (340 Ma, pole MES). The average rate of APW is ~1.5°/m.y. This rate represents the drifting speed of West Gondwanaland across the south pole.

This APW path is in sharp contrast to that proposed by *Bachtadse and Briden* [1991], based mainly on African results, and that proposed by *Klootwijk et al.* [1993], based largely on their preliminary results from Australia, which consider that the paleopole moved progessively south across Africa during the Devonian. However, the results from Africa can be reinterpreted within the limits of the given information to conform with the APW path presented in Figure 9. For detailed

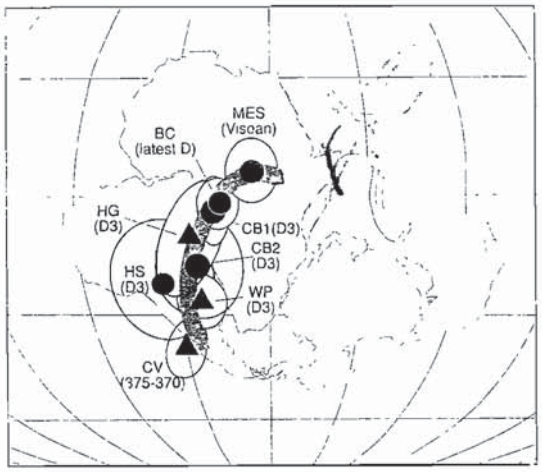


Figure 9. Middle to Late Devonian and Early Carboniferous paleopoles from Australia plotted in Gondwanaland reference (reconstruction parameters are given by *Powell and Li* [1994, Table 1]). Dots are poles from cratonic Australia, and triangles are poles from the Lachlan Fold Belt, southeastern Australia. CV, Comerong Volcanics [*Schmidt et al.*, 1986]; WP, Worange Point Formation [*Thrupp et al.*, 1991]; HS, Hermannsburg Sandstone [*Li et al.*, 1991a]; HG, Hervey Group [*Li et al.*, 1988]; CB1/CB2, Late Devonian Canning Basin limestone of *Hurley and Van der Voo* [1987] and this study; BC, Brewer Conglomerate [*Chen et al.*, 1993]; MES, Mount Eclipse Sandstone [*Li et al.*, 1991b; *Chen et al.*, 1994].

discussion of this aspect, refer to papers by *Chen et al.* [1993] and *Li et al.* [1993a].

In conclusion, this study demonstrates the primary origin of the characteristic remanence in the Canning Basin reef complexes and therefore supports the paleomagnetic interpretation of *Hurley and Van der Voo* [1987] and *Li et al.* [1988] that the Late Devonian paleopole for Gondwanaland lay in Central Africa, thereby implying that there was a large ocean between Laurussia and Gondwanaland in the middle to late Late Devonian. However, paleomagnetic data do permit a close encounter of these two supercontinents during the Early and mid-Devonian [e.g., *Kent and Van der Voo*, 1990; *Li et al.*, 1993b].

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References

Bachtadse, V., and J. C. Briden, Palaeomagnetism of Devonian ring complexes from the Bayuda Desert, Sudan – New constraints on the apparent polar wander path for Gondwanaland, *Geophys. J. Int., 104*, 635-646, 1991.

Balme, B. E., and C. W. Hassell, Upper Devonian spores from the Canning Basin, Western Australia, *Micropaleontology, 8*, 1-28, 1962.

Briden, J. C., and B. A. Duff, Pre-Carboniferous paleomagnetism of Europe, north of the Alpine Orogenic Belt, in *Palaeoreconstruction of the Continents, Geodyn. Ser.*, vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 137-149, AGU, Washington, D.C., 1981.

Chen, Z., Z. X. Li, C. McA. Powell, and B. E. Balme, Palaeomagnetism of the Brewer Conglomerate in central Australia, and fast plate movement during the Late Devonian, *Geophys. J. Int., 115*, 564-574, 1993.

Chen, Z., Z. X. Li, C. McA. Powell, and B. E. Balme, An Early Carboniferous paleomagnetic pole for Gondwanaland: New results from diamond-drill core materials of the Mount Eclipse Sandstone in the Ngalia Basin, central Australia. *J. Geophys. Res., 99* (B2), 2909-2924, 1994.

Coleman, P. J., Atrypa in Western Australia, *J. Paleontol., 25*, 667-690, 1951.

Druce, E. C., Conodont biostratigraphy of the Upper Devonian reef complexes of the Canning Basin, Western Australia, *Bull Bur. Miner. Resour Geol. Geophys. Aust., 158*, 303 pp., 1976.

Embleton, B.J.J., The palaeomagnetism of some Palaeozoic sediments from central Australia, *J. Proc. R. Soc. N. S. W., 105*, 86-93, 1972.

Fisher, R. A.., Dispersion on a sphere, *Proc. R. Soc., London, A, 217*, 295-305, 1953.

Forman, D. J., and D. V. Wales, Geological evolution of the Canning Basin, Western Australia, *Bull. Bur. Miner. Resour. Geol. Geophys. Aust., 210*, 91 pp., 1981.

Glenister, B. F., Upper Devonian ammonoids from the *Manticoceras* Zone, Fitzroy basin, Western Australia, *J. Paleontol., 32*, 58-96, 1958.

Glenister, B. F., and G. J. Klapper, Upper Devonian conodonts from the Canning Basin, Western Australia, *J Paleontol., 40*, 777-842, 1966.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54*, 131-167, 1949.

Grey, K., Devonian spores from the Gogo Formation, Canning Basin, *Annu. Rep. Geol. Surv. West. Aust., 1973*, 96-99, 1974.

Grey, K., Devonian atrypid brachiopods from the reef complexes of the Canning Basin, Rep., *5*, 70 pp., Geol. Surv. of West. Aust., Perth, 1978.

Hailwood, E. A., Paleomagnetism of the Msissi norite (Morocco) and the Paleozoic reconstruction of Gondwanaland, *Earth Planet. Sci. Lett., 23*, 376-386, 1974.

Hill, D., and J. S. Jell, Devonian corals from the Canning Basin, Western Australia, *Bull. West. Aust. Geol. Surv., 36*, 116 pp., 1970.

Hurley, N. F., and K. C. Lohmann, Diagenesis of Devonian reefs in the Oscar Range, Western Australia, paper presented at 12th International Sedimentology Congress, Bur. of Miner. Resour., Canberra, A.C.T., 1986.

Hurley, N. F., and R. Van der Voo, Paleomagnetism of Upper Devonian reefal limestones, Canning Basin, Western Australia, *Geol. Soc. Am. Bull., 98*, 138-146, 1987.

Hurley, N. F., and R. Van der Voo, Magnetostratigraphy, Late Devonian iridium anomaly, and impact hypotheses, *Geology, 18*, 291-194, 1990.

Idnurm, M., and C. A. Heinrich, A palaeomagnetic study of hydrothermal activity and uranium mineralization at Mt Painter, South Australia, *Aust. J. Earth Sci., 40*(1), 87-101, 1993.

Kent, D. V., and S. R May, Polar wander and palaeomagnetic reference pole controversies, *Rev. Geophys., 25*(5), 961-970, 1987.

Kent, D. V., and R. Van der Voo, Palaeozoic palaeogeography from palaeomagnetism of the Atlantic-bordering continents, in Palaeozoic Palaeogeography and Biogeography, edited by W. S. McKerrow and C. R. Scotese, *Mem. 12*, pp. 49-56, Geol. Soc. of London, 1990.

Kerans, C., and P. E. Playford, Scheck breccias from Devonian reef complexes of Canning Basin, Western Australia (abstract), *AAPG Bull., 68*(4), 495, 1984.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J R. Astron. Soc , 62*, 699-718, 1980.

Klootwijk, C. T., J. W. Giddings, and P. Percival, Palaeomagnetic reconnaissance of Upper Palaeozoic volcanic, northeastern Queensland, *Rec. 1993/36*, 88 pp., Bur. of Miner. Resour. Canberra, A.C.T., Australia, 1993.

Li, Z. X., and C. McA Powell, Late Proterozoic to Early Paleozoic paleomagnetism and the formation of Gondwanaland, in *Gondwana 8; Assembly, Evolution and Dispersal*, edited by R. H. Findlay, M. R. Banks, J. J. Veevers, and R. Unrug, pp. 9-21, A. A. Balkema, Rotterdam, Netherlands, 1993.

Li, Z. X., P. W. Schmidt, and B.J.J. Embleton, Palaeomagnetism of the Hervey Group, central New South Wales and its tectonic implications, *Tectonics, 7*, 351-367, 1988.

Li, Z. X., C. McA. Powell, and P. W. Schmidt, Syn-deformational remanent magnetization of the Mount Eclipse Sandstone, central Australia, *Geophys. J. Int., 99*, 205-222, 1989.

Li, Z. X., C. McA. Powell, G. A. Thrupp, and P. W. Schmidt, Australian Palaeozoic palaeomagnetism and tectonics, II, A revised apparent polar wander path and palaeogeography, *J. Struct. Geol., 12*(5/6), 567-575, 1990.

Li, Z. X., C. McA. Powell, B. J. J. Embleton, and P. W. Schmidt, New palaeomagnetic results from the Amadeus Basin and their implications for stratigraphy and tectonics, in Geological and Geophysical Studies of the Amadeus Basin, Central Australia, edited by R. J. Korsch and J. M. Kennard, *Bull. Bur. Miner. Resour. Geol. Geophys. Aust., 236*, 349-360, 1991a.

Li, Z. X., C. McA. Powell, and D. G. Morris, Syn-deformational and drilling-induced remanent magnetizations from diamond drill cores of the Mt Eclipse Sandstone, central Australia, *Aust. J. Earth Sci., 38*(4), 473-484, 1991b.

Li, Z. X., Z. Chen, and C. McA. Powell, New Late Palaeozoic palaeomagnetic results from Australia, and revision of the Gondwanan apparent polar wander path, *Explor. Geophys., 24*, 263-268, 1993a.

Li, Z. X., C. McA. Powell, and A. Trench, Palaeozoic global reconstructions, in *Palaeozoic Vertebrate Biostratigraphy and Biogeography*, edited by J. A. Long, pp. 25-53, Belhaven, London, 1993b.

Long, J. A., Late Devonian fishes from Gogo, Western Australia, *Nat. Geogr. Res., 4*(4), 436-450, 1988.

McFadden, P. L., and M. W. McElhinny, Classification of the reversal test in palaeomagnetism, *Geophys. J. Int., 103*, 725-729, 1990.

Meert, J., R. Van der Voo, C. McA. Powell, Z. X. Li, M. W. McElhinny, Z. Chen, and D.T.A. Symons, Is there a plate-tectonic speed limit? *Nature, 363*, 216-217, 1993.

Nicoll, R. S., and E. C. Druce, Conodonts from the Fairfield Group, Canning Basin, Western Australia, *Bull. Bur. Miner. Resour. Geol. Geophys. Aust., 190*, 134 pp., 1979.

Petersen, M.S., Upper Devonian (Fammenian) ammonoids from the Canning Basin, Western Australia, *Mem. 8*, (Supplement to *J. Paleontol., 49*, No. 5), 55 pp., The Society of Economic Paleontologists and Mineralogists, The Paleontological Society, Tulsa, Okla., 1975.

Playford, G., Plant microfossils from the Upper Devonian and Lower Carboniferous of the Canning Basin, Western Australia, *Palaeontographica, 158* (B), 1-71, 1976.

Playford, P. E., Devonian "Great Barrier Reef" of Canning Basin, Western Australia, *AAPG Bull., 64*(6), 814-840, 1980.

Playford, P. E., Devonian reef complexes of the Canning Basin, Western Australia, *Field Excursion Guidebook*, Fifth Australian Geological Convention, 64 pp., Geological Society of Australia, Perth, Western Australia, 1981.

Playford, P. E., Platform-margin and marginal-slope relations in Devonian reef complexes of the Canning Basin, in *The Canning Basin, W.A.*, edited by P. G. Purcell, pp. 189-214, Geological Society of Australia and Petroleum Exploration Society of Australia, Perth, Western Australia, 1984.

Playford, P. E., and A. E. Cockbain, Algal stromatolites: Deepwater forms in the Devonian of Western Australia, *Science, 165*, 1008-1010, 1969.

Playford, P. E., A. E. Cockbain, E. C. Druce, and J. L. Wray, Devonian stromatolites from the Canning Basin, Western Australia, in *Stromatolites, Dev. Sedimentol.*, vol. 20, pp. 543-563, Elsevier, New York, 1976.

Powell, C. McA., and Z. X. Li, Gondwanaland reconstruction, in Permian–Triassic Pangean basins and fold belts along the Panthalassan margin of Gondwanaland, edited by J. J. Veevers and C. McA. Powell, *Mem. Geol Soc. Am., 184*, in press, 1994.

Ross, J. P., Ordovician, Silurian, and Devonian Bryozoa of Australia, *BMR Bull., 50*, 172 pp., 1961.

Salmon, E., R. Montigny, J. B. Edel, A. Pique, R. Thuizat, and M. Westphal, The Msissi norite revisited: K/Ar dating, petrography and paleomagnetism, *Geophys. Res. Lett., 13*(8), 741-743, 1986.

Schmidt, P. W., Linearity spectrum analysis of multi-component magnetizations and its application to some igneous rocks from south-eastern Australia, *Geophys. J. R. Astron. Soc., 70*, 647-665, 1982.

Schmidt, P. W., B. J. J. Embleton, T. J. Cudahy, and C. McA. Powell, Prefolding and pre-megakinking magnetization from the Devonian Comerong Volcanics, New South Wales, Australia, and their bearing on the Gondwana Pole Path, *Tectonics, 5*, 135-150, 1986.

Schmidt, P. W., B J. J. Embleton, and H. C. Palmer, Pre and post folding magnetization from the Devonian Snowy River Volcanics and Buchan Caves Limestone, Victoria, *Geophys. J. R. Astron. Soc., 91*, 155-170, 1987.

Schmidt, P. W., C. McA. Powell, Z. X. Li, and G. A. Thrupp, Reliability of Palaeozoic palaeomagnetic poles and APWP of Gondwanaland, *Tectonophysics, 184*, 87-100, 1990.

Seddon, G., Frasnian conodonts from the Sadler ridge–Bugle Gap area, Canning Basin, Western Australia, *J. Geol. Soc. Aust., 16*, 723-753, 1970.

Thrupp, G. A., D. V. Kent, P. W. Schmidt, and C. McA. Powell, Palaeomagnetism of red beds of the Late Devonian Worange Point Formation, SE Australia, *Geophys. J. Int., 104*, 179-201, 1991.

Towner, R. R., and D. L. Gibson, Geology of the onshore Canning Basin, Western Australia, *Bull. Bur. Miner. Resour. Geol. Geophys. Aust., 215*, 51 pp., 1983.

Van der Voo, R., The reliability of paleomagnetic data, *Tectonophysics, 184*, 1-9, 1990.

Veevers, J. J., Devonian brachiopods from the Fitzroy basin, Western Australia, *Bull. Bur. Miner. Resour. Geol. Geophys. Aust., 45*, 173 pp., 1959.

Wray, J. L., Upper Devonian calcareous algae from the Canning Basin, Western Australia, *Prof. Contrib., 3*, 76 pp., Color. School of Mines, Golden, 1967.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Palaeomagnetism*, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

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